20/8/26 Exp: 6 A program to implement error detection and correction using Hamming code.

Aim: Write a program to implement error detection and correction using Hamming code concept. Make a run to input data stream and verify error correction feature.

create sender program with below features:

1 Input to sender file should be a text of any length. Program should convert the text to binary.

2 Apply hamming code code concept on binary data and add redundant bits to it.

3 Spave this output in a file called channel.

1) Receiver program should read the input form channel file.

2) Apply hamming code on the binary data to check for errors.

3) If there is an error, display the position of the error.

else somewhere remove the redundant bits and convert the binary data to ascii and display output.

CODE:

Sender program:

```
8 = input ("Enter text:")
b = ''.join (format ord (c), '08b') for c in 8)
m = len (b)
r = 0
while 2**r < m+r+1:
        r+=1
a = [0] * (m+r+1)
j = 0
for i in range (1, m+r+1):
  if i & (i-1):
     a[i] = b[j]
     j+=1
for p in range (r):
   x = 2**p
   a[x] = str (sum (int (a[i] for i
in range (1, len (a)) if
i & x) % 2)
code = ''.join (a[1:])
print ("Hamming code:", code)
open ("channel.txt", "w"). write(code)
```

Recver program:

```
c = list (open ("channel.txt").read ())
n = len (c)
error = 0
p = 1
while p <= n:
parity = sum (int (c[i-1]) for i in range
 if i & p) % 2
 if parity:
      error += p
      p *= 2
 if error:
   print ("error at position:", error)
c[error-1] = '1' if c[error-1] == '0' else '0'
 print ("error corrected")
else:
print ("No error")
data = ''.join (c[i-1] for i in range
(1, n+1) if i & (i-1))
out = ''.join (chr (int (data[i:i+8], 2)
 for i in out (0, len (data), 8))
print ("output:", out)
```

Sample input:

HELLO

Sample Output:

Running code: 1001100000310
No error
Output: HELLO

Result: Thus the code is executed and verified